

2. Hyperphosphorylation of the src kinase hck induced by FcγRI engagement

To assay for src kinase activation following FcγRI engagement, we used a panel of rabbit anti-peptide antisera generated in our laboratory that react specifically with different src family protein tyrosine kinases, including hck, fgr, lyn, and src, all of which are expressed in monocytes. We made use of a sensitive in-situ kinase assay to detect autophosphorylation of src PTK, which accompanies their activation. THP-1 cells were stimulated with anti-FcγRI mAb 197, then permeabilized with α-lysophosphatidylcholine and incubated on ice in the presence of γ -ATP. Cells were then subjected to detergent lysis and immunoprecipitation with the different anti-src antisera. As shown in Figure 2, we detected an increase in phosphorylation in hck in THP-1 cells within 30 seconds of stimulation by anti-FcγRI mAb. In contrast, we did not detect increases in phosphorylation of other src PTK (not shown). These results suggest that activation of the PTK hck is coupled to FcγRI engagement. We are currently evaluating the physical association of src kinases with FcγRI. Our preliminary data suggest that hck and lyn are present in anti-FcγRI immunoprecipitates from THP-1 cells and peripheral blood monocytes, as detected by immunoblotting.

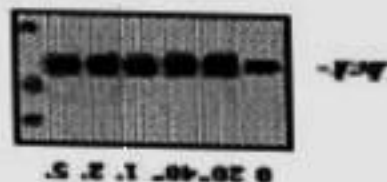


Figure 2. (Continued attached). Hyperphosphorylation of the src kinase hck induced by FcγRI engagement in THP-1 cells. In situ kinase assay was performed as described in text. Anti-hck immunoprecipitates from lysates of cells stimulated for between 0-5 minutes (as indicated) were analyzed by gel electrophoresis and autoradiography. M.W. markers (right hand lane) are at 92, 69 and 46 kDa. Four other experiments gave similar results.

I trust that this information will be helpful to the reviewers of the application.

Sincerely yours,

Carl S. Gheba, M.D.